



# Abdulmalik Abdulmawla

**SPATIAL-DATA TOOLS: ARCHITECTURE, URBAN AND RURAL SPACE, MOBILITY, CITY ECONOMY**

## PROFILE

I am interested in solving spatial problems, in architecture, urban and rural space, economics and mobility. I believe that any complex challenge, and any handling of data, requires intelligently breaking the problem down to its core parameters and simplifying each element until a humanly manageable scale is reached. This is how nature works: complex systems are not made of complex parts, but of very simple ones, intelligently put together.

I come from research and have been building tools for the design, generation and analysis of spatial data for many years. Technology is one of my strengths; my main one is computational thinking about spatial data problems: I break processes down into their elemental parts, translate them into parametric, interactive environments and work my way into the field the data serves, from simple to complex objectives. In doing so I work on the systems behind the tools, on data models, methods and interfaces, with state-of-the-art technologies: from agent-based simulation to AI-assisted, multi-agent coding. My standard is the best balance between user experience and advanced scientific analysis methods; I see early where the users of a tool will stumble.

## EXPERIENCE

**Dec 2023 – Sep 2026**

### Research Associate, Chair of Transport System Planning

Bauhaus-Universität Weimar · Bauhaus.Mobility / European Digital Innovation Hub Mid Germany (formerly EDIH Thuringia)

Design and development of interactive geodata platforms, among them the browser-based analysis platform for Germany's official accident data: sole developer from the data model through the open-data pipelines to the WebGL map, with continuous integration and versioned releases; the official evaluation method M-Uko 2012 as an interactive tool. Data-science consulting for public institutions, training public administrations in open geodata and European data standards, support for dataset publication on the Mobilithek. Supervision of student projects and theses (Digital Transport Research Lab).

**May 2016 – Mar 2023**

### Research Associate, Computer Science in Architecture (InfAR, now InfAU)

Bauhaus-Universität Weimar

Teaching computational methods for urban planning and analysis; development of design and research tools (DeCodingSpaces Toolbox, mineR); supervision of master's theses; conference papers (Space Syntax Symposium, eCAADe) and support of publications, exhibitions and study excursions.

**Dec 2017 – Jan 2019**

### Project Coordinator, DAAD exchange project "Discovering Cities"

Bauhaus-Universität Weimar and German-Jordanian University, Amman · held concurrently with the InfAR post

Wrote the project proposal, managed the project finances, taught the seminar in Weimar and prepared the workshop in Amman.

**May 2016 – 2020**

### Contributing Author, DeCodingSpaces Toolbox

DecodingSpaces GbR, Weimar

C# development for a Grasshopper toolbox for urban analysis; testing, documentation and consulting, among them an interactive masterplan-analysis tool for the Blankenburger Süden development (client: TSPA, Berlin, 2019).

## CONTACT

### LOCATION

Weimar, Germany

### EMAIL

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### PORTFOLIO

abdulmalikabdulmawla.github.io

### GITHUB

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### ORCID

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## LANGUAGES

**Arabic** · native

**English** · fluent

**German** · very good

## SKILLS

### Spatial data and analysis

Official German geodata (ALKIS, cadastre, LoD2/CityGML, WFS/OGC), OpenStreetMap, spatial SQL (DuckDB, GeoParquet), QGIS, network analysis, space syntax, statistics (R)

### Tool and application development

Python, TypeScript/JavaScript (React, deck.gl), C# (Grasshopper), simulation and agent-based models, Git, continuous integration, AI-assisted development (Claude Code)

### Teaching and consulting

Seminars, workshops, training public administrations in working with open geodata

### Design and visualisation

Rhino, AutoCAD, Revit, Adobe Photoshop, Illustrator, InDesign

## TEACHING

- 2023 – 2026** · Digital Traffic Simulation Lab · Open Mobility Data · Rent-a-Data-Scientist, Chair of Transport System Planning, Bauhaus-Universität Weimar
- 2016 – 2023** · Algorithmic Architecture · Parametric Urban Design and Analysis · SYNCITY I and II · Circular Urbanism, InfAR/InfAU, Bauhaus-Universität Weimar
- 2018** · Discovering Cities teaching seminar (DAAD), Bauhaus-Universität Weimar
- 2014 – 2016** · Design 5 Studio · Web Application in Architecture, AASTMT, Cairo
- Supervision of master's theses and student projects, 2016 – 2023 at InfAR/InfAU and 2023 – 2026 in the Digital Transport Research Lab

## WORKSHOPS (ORGANISER)

- 2018** · Discovering Cities Workshop, German-Jordanian University, Amman
- 2018** · Urban Analysis, Synthesis and Exploration with Grasshopper, eCAADe 2018, Łódź
- 2016** · Rural-to-Urban Transformation Workshop, Emerging City Lab, Addis Ababa
- 2015** · Simulating Agent-Based Systems using Processing, Cairo

## RECOGNITION

- 2026** · Ten-year service recognition at the Bauhaus-Universität Weimar, presented by the University President

## COMPETITIONS

- 2014** · Lusail Underpasses Public Art Competition, phases 1 and 2, Doha
- 2007** · 2nd place, 4th National Architecture Students Competition, Cairo University
- 2006** · 1st place, 3rd National Architecture Students Competition, Cairo University

## EXPERIENCE (CONTINUED)

**Oct 2014 – Apr 2016**

### Research Associate, Teaching and Computational Design

Arab Academy for Science, Technology and Maritime Transport (AASTMT), Cairo

Teaching computational and parametric design (Grasshopper, Processing, Arduino) in a RIBA Part I certified programme.

**Jan 2012 – Apr 2016**

### Co-founder and Architect, Gozour Studio

Mansoura, Egypt

Design studio and architecture consultancy: residential, education and competition projects, among them the Lusail Underpass Competition, Qatar.

**May 2009 – Sep 2011**

### Junior Architect, EAI Group, Abu Dhabi

Elite Design and Engineering Consultancy, semi-government: client representative for outsourced projects, client consultant for in-house projects

**Jul 2008 – Apr 2009**

### Freelance Architect, Egypt

3D modelling and visualisation for residential construction projects

**Jul 2007 – Jun 2008**

### Junior Architect, Invert Studios, Cairo

Architectural design and planning at the office of Kamel Louqman and Amir Wahid

**Jun – Aug 2005**

### Intern, Al Turath Engineering Consultancy, Abu Dhabi

CAD design, layout and plotting of complete project sets

## EDUCATION

**Nov 2018 – present**

### PhD candidate, in progress · Computer Science in Architecture and Urbanism (InfAU), Bauhaus-Universität Weimar

Working title: "Spatial dependency of urban retail: detailed methods for positioning street-level activities" · supervisor: Prof. Dr. Reinhard König

**2013**

### Master of Arts in Architecture · Hochschule Anhalt, Dessau-Roßlau, Germany

Thesis: "Heterogeneous-Atelier Industry" (with Ali Farhan): the informal atelier networks of Şişhane, Istanbul, mapped and simulated as one bottom-up factory

**2007**

### BSc Architecture and Engineering · Faculty of Engineering, Mansoura University, Egypt

## SELECTED PUBLICATIONS

- 2022** **Abdulmawla, A.**, Bielik, M., Schneider, S., König, R. Mapping active frontages: a method for linking street-level activities to their surrounding urban forms. A case study of the city of Weimar, Germany. *13th International Space Syntax Symposium*, Bergen.
- 2022** **Abdulmawla, A.**, Schneider, S., König, R., Bielik, M., Fuchkina, E. Parametric urban data structuring and spatial query: advanced data mapping and selection methods for parametric modelling environments. *40th eCAADe*.
- 2019** Bielik, M., König, R., Fuchkina, E., Schneider, S., **Abdulmawla, A.** Evolving configurational properties: simulating multiplier effects between land use and movement patterns. *12th International Space Syntax Symposium*, Beijing.
- 2018** **Abdulmawla, A.**, Schneider, S., Bielik, M., König, R. Integrated data analysis for parametric design environments: mineR, a Grasshopper plugin based on R. *36th eCAADe*.
- 2017** König, R., Bielik, M., Knecht, K., **Abdulmawla, A.**, Fuchkina, E. New methods for urban analysis and simulation with Grasshopper: using the DeCodingSpaces-Toolbox. *CAAD Futures 2017*, Istanbul.
- 2017** Schneider, S., **Abdulmawla, A.**, Donath, D. The effect of the street network on movement patterns and land use in small cities: a comparative study of three 10K cities in Thuringia, Germany. *11th International Space Syntax Symposium*, Lisbon.
- 2017** Denmark, M., Schneider, S., König, R., **Abdulmawla, A.**, Donath, D. Towards a modular design strategy for urban masterplanning: experiences from a parametric urban design studio on emerging cities in Ethiopia. *eCAADe 2017*, Rome.
- 2016** **Abdulmawla, A.**, Farhan, A. Heterogeneous assemblages of atelier industries. *SimAUD 2016*, London (poster).

Full list: [researchgate.net/profile/Abdulmalik-Abdulmawla](https://researchgate.net/profile/Abdulmalik-Abdulmawla)